LAB 9

Numerical Integration

Assignment:

Assignment -

One numerical as suggested by Faculty.

Sample numerical.

19.4 Evaluate the following integral:

$$\int_{-2}^{4} (1 - x - 4x^3 + 2x^5)\,dx$$

(a) analytically, **(b)** single application of the trapezoidal rule, **(c)** composite trapezoidal rule with $n = 2$ and 4, **(d)** single application of Simpson's 1/3 rule, **(e)** Simpson's 3/8 rule, and

Hand Calculation:

(a) Analytics method:

$$\int_{-2}^{4} (1 - x - 4x^3 + 2x^5)\,dx$$

$$\Rightarrow \left[x - x^2/2 - x^4 + x^6/3\right]_{4}^{-2}$$

(b) Trapazodial Rule:

$$I = \frac{b-a}{2}\left[f(a) + f(b)\right]$$

$f(-2) = -2$ $\qquad$ $f(4) = 1669$

(c) Trapazoidal Rule $(n=2)$

$$I = h/2\,(f_0 + 2f_1 + f_1)$$

(d) Trapazoidal Rule ($n=4$)

$$I = h/2\,(f_0 + 2(f_1+f_2+f_3) + f_4)$$

(e) Simpson's 1/3 rule:

$$I = h/3\,[f_0 + 4f_1 + f_2]$$

(f) Simpson's 3/8 rule:

$$I = 3h/8\,[f_0 + 3f_1 + 3f_2 + f_3]$$

Matlab Code

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```
clc
a = -2;
b = 4;
%Analytical
% F(x) = x - x^2/2 - x^4 + x^6/3
Fa = a - (a^2)/2 - (a^4) + (a^6)/3;
Fb = b - (b^2)/2 - (b^4) + (b^6)/3;
I1 = Fb - Fa;

fprintf('Analytical   = %.2f\n', I1);
```

Command Window

```
Analytical   = 1104.00
```

Figure 1 Analytical Method

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```
clc
%trapazoidal

fa = 1 - a - 4*a^3 + 2*a^5;
fb = 1 - b - 4*b^3 + 2*b^5;

I2 = ((b - a)/2) * (fa + fb);
fprintf('Trap single  = %.2f\n', I2);
```

Command Window

```
Trap single  = 5280.00
```

Figure 2 Trapezoidal Rule

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```
clc
% composite trapazoidal
n_values = [2 4];
for k = 1:length(n_values)
    n = n_values(k);
    h = (b - a)/n;
    sum = 0;

    for i = 1:n-1
        xi = a + i*h;
        fi = 1 - xi - 4*xi^3 + 2*xi^5;
        sum = sum + fi;
    end
    f0 = 1 - a - 4*a^3 + 2*a^5;
    fn = 1 - b - 4*b^3 + 2*b^5;
    I_trap(k) = (h/2) * (f0 + 2*sum + fn);
end
I3 = I_trap(1); % n = 2
I4 = I_trap(2); % n = 4
fprintf('Trap n=2     = %.2f\n', I3);
fprintf('Trap n=4     = %.2f\n', I4);
```

Command Window

```
Trap single  = 5280.00
```

Figure 3 Trapezoidal Rule

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```
clc
%simpson 1/3
h = (b - a)/2;
x1 = (a + b)/2;
f0 = 1 - a - 4*a^3 + 2*a^5;
f1 = 1 - x1 - 4*x1^3 + 2*x1^5;
f2 = 1 - b - 4*b^3 + 2*b^5;
I5 = (h/3) * (f0 + 4*f1 + f2);
fprintf('Simpson 1/3  = %.2f\n', I5);
```

Command Window

```
Simpson 1/3  = 1752.00
```

Figure 4 Simpson 1/3 rule

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```
clc
a = -2;
b = 4;
%simpson 3/8
n = 3;
h = (b - a)/n;
sum = 0;
for i = 1:n-1
    xi = a + i*h;
    fi = 1 - xi - 4*xi^3 + 2*xi^5;
    sum = sum + 3*fi;
end
f0 = 1 - a - 4*a^3 + 2*a^5;
fn = 1 - b - 4*b^3 + 2*b^5;

I6 = (3*h/8) * (f0 + sum + fn);
fprintf('Simpson 3/8  = %.2f\n', I6);
```

Command Window

```
Simpson 3/8  = 1392.00
```

Figure 5 Simpson 3/8 Rule

Conclusion:

Here the analytical method gives the exact value of the integral while other numerical method provides approximate result. Also increase in the number of intervals also improves the result.